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Methodological Advances

Comparison of Critical Force to EMG Fatigue Thresholds during Isometric Leg Extension

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ABSTRACT

HENDRIX, C. R., T. J. HOUSH, G. O. JOHNSON, M. MIELKE, C. L. CAMIC, J. M. ZUNIGA, and R. J. SCHMIDT. Comparison of Critical Force to EMG Fatigue Thresholds during Isometric Leg Extension. *Med. Sci. Sports Exerc.*, Vol. 41, No. 4, pp. 956–964, 2009. Theoretically, the critical force (CF) and the EMG fatigue threshold (EMG_{FT}) tests demarcate fatiguing from nonfatiguing isometric torque levels. **Purpose**: The purpose of this study was twofold: 1) to determine whether the mathematical model for estimating the EMG_{FT} during cycle ergometry was applicable to isometric leg extension muscle actions and 2) to compare the mean torque level from the CF test to those of EMG_{FT} tests for the vastus lateralis (VL), vastus medialis (VM), and rectus femoris (RF) muscles during isometric muscle actions. **Methods**: The slope coefficient of the linear relationship between total "isometric work" (W_{lim} in newton-meters per second) and time to exhaustion (T_{lim} in seconds) was defined as the CF. The EMG_{FT} was defined as the *y*-intercept of the isometric torque versus EMG fatigue curve slope coefficient relationship. **Results**: There was a significant ($P < 0.05$) mean difference between the CF (25.9 ± 12.1 N·m) and the EMG_{FT} value for the RF (41.1 ± 20.7 N·m) muscle. There were no significant differences, however, in EMG_{FT} values among the three superficial muscles of the quadriceps femoris. In addition, the mean CF (17.6% maximum voluntary isometric contraction [MVIC]) occurred at a percentage of MVIC that is typically not affected by circulatory occlusion (20% MVIC), whereas the mean EMG_{FT} values for the VL (25.9% MVIC), VM (22.9% MVIC), and RF (27.8% MVIC) exceeded this threshold. **Key Words:** ANAEROBIC WORK CAPACITY, ISOMETRIC WORK (W_{lim}), LIMIT TIME (T_{lim}), EMG, EMG AMPLITUDE

The critical force (CF) test (27), theoretically, provides approximations of two parameters: the CF and the anaerobic work capacity (AWC). CF is the maximal isometric force (or torque) that a muscle or a muscle group can maintain for an extended duration without fatigue and is related to the circulatory conditions of the muscle. AWC is defined as the total "isometric work" that can be achieved using only energy reserves within the muscle such as adenosine triphosphate (ATP), phosphocreatine, glycogen, and oxygen bound to myoglobin (7,27,29). Although, by definition, no mechanical work is accomplished during an isometric muscle action, Monod and Scherrer (27) suggested that an appreciation of the fatiguing nature of isometric muscle actions is found in the time to exhaustion for a given force or torque level. Thus, to derive the CF and the AWC from the CF test, Monod and Scherrer (27) used an isometric analogue of mechanical work called the limit work (W_{lim}) that is calculated as the force (F) of the isometric muscle action multiplied by the time to exhaustion (or limit time, T_{lim}) that the F can be maintained. That is,

$$W_{lim} = FT_{lim}, \quad [1]$$

where W_{lim} is the total amount of "isometric work" that can be accomplished at a specific level of F; F is the level of isometric force of the muscle action; and T_{lim} is the time to exhaustion at the specific F. On the basis of these definitions, the CF test involves a series of fatiguing, isometric muscle actions (three or four) at different F levels that are maintained continuously for as long as possible. The W_{lim} (FT_{lim}) is then calculated for each F and plotted as a function of the T_{lim} values for the muscle actions (Fig. 1). The linear relationship between W_{lim} and T_{lim},

$$W_{lim} = \mathrm{AWC} + \mathrm{CF}(T_{lim}), \quad [2]$$

is then used to estimate the CF and the AWC values. Theoretically, the slope of the W_{lim} versus T_{lim} relationship

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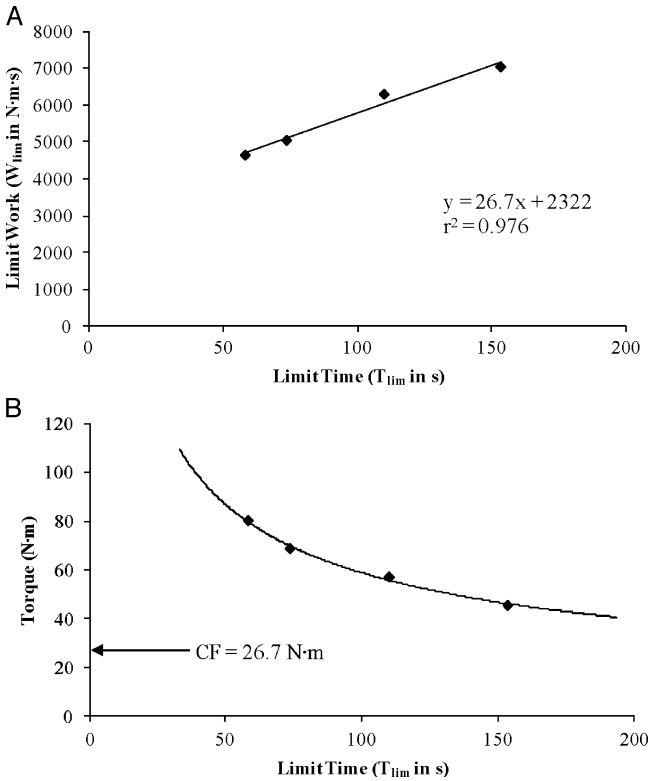

FIGURE 1—Method for estimating the critical force (CF) from data for subject 3. The slope coefficient of the W_{lim} versus T_{lim} relationship is the CF (26.7 N·m) (A), which also represents the asymptote of the torque versus T_{lim} relationship (B).

(expressed in units of F) is the CF and describes the F that corresponds to the asymptote of the F versus T_{lim} relationship (Fig. 1). The y-intercept of the W_{lim} versus T_{lim} relationship (expressed in units of F·s) is defined as the AWC. Thus, the CF is the isometric F threshold above which fatigue will occur during a sustained muscle action at a predictable T_{lim} but below which "…exhaustion cannot occur" (27, p. 332).

Numerous studies (3,12,14,28) have examined the EMG amplitude versus time relationship during a fatiguing task. For example, deVries (12) reported that the EMG amplitude versus time relationships during fatiguing isometric muscle actions were linear, and the slope coefficients were directly proportional to the imposed F but inversely proportional to the T_{lim}. Based on these findings, deVries et al. (13) proposed the EMG fatigue threshold (EMG_{FT}) test for cycle ergometry, which estimates the power output associated with the onset of neuromuscular fatigue. The EMG_{FT} test for cycle ergometry involves the determination of the rate of rise in EMG amplitude as a function of time (slope coefficient) at three or four fatiguing power outputs (Fig. 2A). The four power outputs are then plotted as a function of their corresponding slope coefficients for the EMG amplitude versus time relationships, and the EMG_{FT} is defined as the y-intercept (Fig. 2B) (13). Therefore, the EMG_{FT} test (13) estimates the highest power output that can be maintained without neuromuscular evidence of fatigue (i.e., a slope coefficient for the EMG amplitude vs time relationship of zero) (26,30). The mathematical model of the EMG_{FT} test proposed by deVries et al. (13) for cycle ergometry, however, has not been applied to isometric muscle actions. If applicable, the EMG_{FT} test would provide an estimate of the highest level of isometric force (or torque) that can be maintained for an extended period without fatigue.

Both the CF and the EMG_{FT} tests, in theory, estimate "fatigue thresholds" that demarcate fatiguing from nonfatiguing isometric F levels. The tests (CF and EMG_{FT}), however, use different methodologies (the EMG_{FT} test measures neuromuscular activity, whereas the CF test describes the force vs duration relationship). No previous studies have compared the F levels derived from the CF and the EMG_{FT} tests of the leg extensors. Therefore, the purpose of this study was twofold: 1) to determine whether the mathematical model of deVries et al. (13) for estimating the EMG_{FT} during cycle ergometry was applicable to isometric leg extension muscle actions and 2) to compare the mean torque level from the CF test to those of EMG_{FT} tests for the vastus lateralis (VL), the vastus medialis (VM), and the rectus femoris (RF) muscles during isometric muscle actions. Based on the findings of deVries et al. (13), we hypothesized that 1) the EMG_{FT} cycle ergometry test (13) would be applicable to isometric muscle actions of the leg extensors and 2) the mean torque level for CF would be less than those at the EMG_{FT} for the VL, the VM, and the RF muscles.

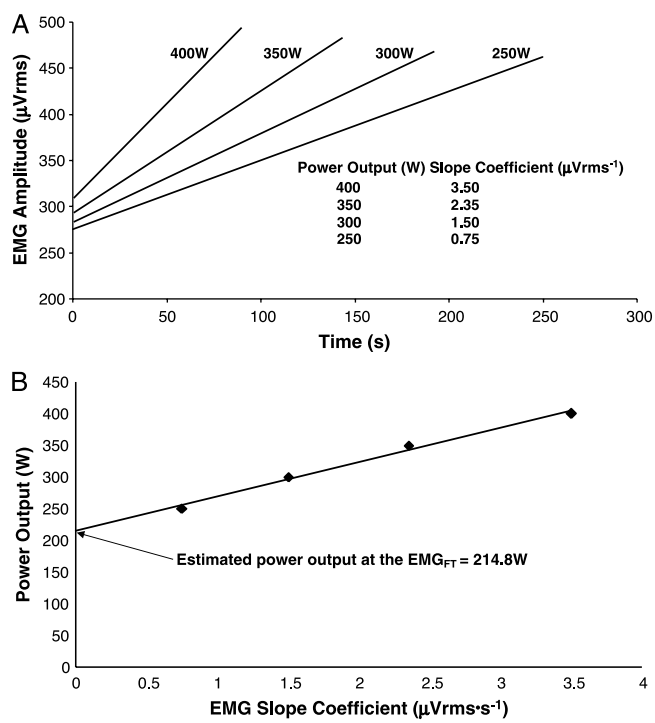

FIGURE 2—Theoretical model for estimating EMG_{FT} test during cycle ergometry. A. The EMG amplitude (µVrms) data are plotted as a function of time (s) during the four fatiguing power outputs on a cycle ergometer (e.g., 250, 300, 350, and 400 W). B. The slope coefficients from panel A are plotted for each of the four power outputs, and the EMG_{FT} was estimated as the y-intercept value (214.8 W).

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METHODS

Subjects. Eight adults (four men and four women; mean ± SD: age = 21.5 ± 1.3 yr; height = 169.9 ± 16.3 cm; weight = 69.6 ± 19.8 kg) volunteered for this study. None of the subjects were competitive athletes, but they participated regularly in walking (n = 1), bicycling (n = 2), jogging or running (n = 8), weight training (n = 3), and/or various sports (n = 1; basketball or boxing). All procedures were approved by the university institutional review board for human subjects, and each subject completed a health history questionnaire and gave their written informed consent to participate before testing. The subjects visited the laboratory on five occasions. The first visit served as an orientation to familiarize the subject with the testing protocols and equipment used during the study. During the second visit, maximum voluntary isometric contraction (MVIC) was determined, and one continuous, fatiguing muscle action was performed to determine the T_{lim} at a randomly ordered percentage of MVIC (approximately 30%, 45%, 60%, or 75% ± 5%). The subsequent three visits were to determine the T_{lim} values for the remaining randomly ordered percentages of MVIC. Each laboratory visit was separated by 24 h and was performed at the same time of day.

Determination of MVIC. All isometric leg extension testing, performed by the dominant leg (based on kicking preference), began with a warm-up of five, 6-s submaximal isometric muscle actions. The subjects were instructed to provide an effort corresponding to approximately 50% of their MVIC. After the warm-ups and the 2-min rest, two MVIC trials were performed. Each MVIC trial consisted of a 6-s maximal isometric muscle action at a joint angle of 120° between the thigh and the leg. Peak torque was recorded by a calibrated Cybex II isokinetic dynamometer. The subject rested 2 min between MVIC trials. The highest measured peak torque level was used as the individual's MVIC. After the second MVIC trial, each subject rested 5 min, then performed the first continuous, isometric muscle action at a randomly determined percentage of MVIC to determine the T_{lim}.

Determination of CF. The determination of CF was based on the work of Monod and Sherrer (27) and involved continuous, fatiguing, isometric muscle actions of the leg extensors at three or four (separated by a minimum of 24–48 h) randomly ordered percentages of MVIC at approximately 30%, 45%, 60%, and 75% (±5%). During visits 2–5, each subject performed one continuous, fatiguing, isometric muscle action to voluntary exhaustion to determine the T_{lim}. We defined exhaustion as the inability to maintain the assigned torque level (i.e., a drop in torque greater than 5% of MVIC). Throughout each muscle action, the subject was given verbal feedback to produce the desired amount of torque. Although isometric muscle actions do not produce mechanical work, metabolic work is performed (27). Thus, based on the method of Monod and Scherrer (27), we calculated the W_{lim} for each isometric muscle action by multiplying the torque of the muscle action by the T_{lim} (equation 1). The W_{lim} values from the continuous, fatiguing, isometric muscle actions were plotted as a linear function of the corresponding T_{lim} values (equation 2; Fig. 1) (27). The CF was defined as the slope coefficient of the W_{lim} versus T_{lim} relationship expressed in newton meters (Fig. 1).

Determination of EMG_{FT}. EMG were recorded during each of the continuous, fatiguing, isometric muscle actions (visits 2–5). Three separate bipolar surface (3.0 cm center-to-center) electrode (Quinton Quick prep silver/silver oxide) arrangements were placed over the superficial muscles of the quadriceps femoris of the dominant thigh (Fig. 3) according to the recommendations of SENIAM 8 (20). The electrodes for the vastus lateralis (VL) muscle were placed over the lateral portion of the muscle approximately 33% of the distance between the superior, lateral border of the patella to the anterior superior iliac spine (ASIS). With the subject in the standing position and the dominant leg fully extended, a reference line was drawn from the ASIS to the superior lateral border of the patella to identify the VL electrode-placement site (37). The electrode-placement sites were located 4–5 cm lateral to the reference line (25). A standard goniometer (Smith & Nephew Rolyan, Inc., Menomonee Falls, WI) was used to orient the electrodes at a 20° angle to the reference line to approximate the pennation angle of the VL muscle (Fig. 3) (1,18). A reference line was drawn to place the electrodes for the vastus medialis (VM) muscle approximately 20% of the distance between the medial knee joint line and the ASIS. A standard goniometer was used to orient the electrodes at a 50° angle to the reference line to approximate the pennation angle of the VM (Fig. 3) (37). A reference line was drawn to place the electrodes for the rectus femoris (RF) muscle approximately 50% of the distance between the superior border of the patella and the inguinal crease (10,38) (Fig. 3). The reference electrodes were placed over the iliac crest. Interelectrode impedance was kept below 2000 Ω by shaving the area and by careful skin abrasion. The EMG signal was preamplified (gain 1000×) using a differential amplifier (BIOPAC Systems Inc., Santa Barbara, CA; bandwidth 10–500 Hz). Six, 5-s epochs (1–5, 11–15, 21–25, 31–35, 41–45, and 51–55 s) of the EMG signal were collected during each minute of the muscle action. The raw EMG signals were digitized at 1000 Hz and stored in a personal computer (Macintosh 7100/80 AV Power PC; Apple Computer, Inc., Cupertino, CA) for subsequent analyses. All signal processing was performed using custom programs written with LabVIEW programming software (version 7.1; National Instruments, Austin, TX). The EMG signals were digitally band-pass filtered (fourth-order Butterworth) at 10–500 Hz. The EMG amplitude (microvolts root mean square [μVrms]) values were calculated for each trial. The rate of rise in EMG amplitude as a function of time (slope coefficient) was calculated for each of the continuous isometric muscle actions for the VL, the VM,

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FIGURE 3—Example of EMG electrode placements.

and the RF muscles for each subject (Fig. 4A). The torque levels were then plotted as a function of the slope coefficients for the EMG versus time relationships for each of the three superficial muscles of the quadriceps femoris (Fig. 4B). The EMG_{FT} was defined as the *y*-intercept of the torque level versus slope coefficient (EMG vs time) plot for each muscle (VL, VM, and RF), resulting in three EMG_{FT} values for each subject (13). Due to technical problems with the EMG equipment, three of the eight subjects completed only three continuous isometric muscle actions.

Statistical analyses. Linear regression was used to estimate the CF (Table 1) and the EMG_{FT} values for the VL, the VM, and the RF muscles (Table 2). It is not uncommon in studies that use a within-subjects design (repeated measures) for there to be missing data points for a variety of reasons (6,9,40). In the present study, this was the case for 2 of the 24 EMG_{FT} determinations (VM for subject 7 and RF for subject 8). Two common procedures for dealing with missing data points are as follows: 1) drop the subject(s) with missing data (6,9,40) or 2) replace the missing values with estimations using a method such as the second approximation procedure of Yates (40). In the present study, Yates' (40) second approximation procedure was used to estimate EMG_{FT} values for the VM for subject 7 and the RF for subject 8 (Table 2). A one-way (1 × 4) repeated-measures ANOVA was used to compare the mean values for CF and the EMG_{FT} for the VL, the VM, and the RF muscles. When appropriate, *post hoc* Bonferroni procedure was used to determine which means were significantly different from each other. An alpha of $P < 0.05$ was considered statistically significant for all analyses. The data were analyzed using the Statistical Package for the Social Sciences software (version 15.0; SPSS Inc., Chicago, IL).

RESULTS

Table 1 includes the data for each subject for the CF test. The mean ± SD values for MVIC and CF were 147.6 ± 87.6 and 25.9 ± 12.1 N·m, respectively. The value for CF represented 17.6% ± 6.2% of MVIC and ranged from 12.7% to 29.1% MVIC. The r^2 values for the W_{lim} versus T_{lim} relationships ranged from 0.706 to 0.994. Table 2 shows the data for each subject and muscle (VL, VM, and RF) for the EMG_{FT} tests. The mean ± SD EMG_{FT} values for the VL, the VM, and the RF muscles were 38.3 ± 22.4, 33.8 ± 22.1, and 41.1 ± 20.7 N·m, respectively. These EMG_{FT} values represented 25.9% ± 8.4%, 22.9% ± 8.8%, and 27.8% ± 8.0% of MVIC, respectively. The r^2 of all three muscles (VL, VM, and RF) for the EMG versus time and the torque versus EMG slope coefficient relationships ranged from 0.010 to 0.993 and from 0.637 to 0.999, respectively. The results indicated that there was a significant mean difference between the CF and the EMG_{FT} for the RF muscle. There were no significant mean differences, however, among the EMG_{FT} values for the three superficial muscles of the quadriceps femoris (VL, VM, and RF). Table 3 provides zero-order correlation matrices among the mean estimated torque values as well as the %MVIC values from the CF test and the EMG_{FT} tests for the VL, the VM, and the RF muscles. The torque values (N·m) from the four fatigue thresholds (CF and EMG_{FT} for the VL, VM, and RF) were significantly intercorrelated at

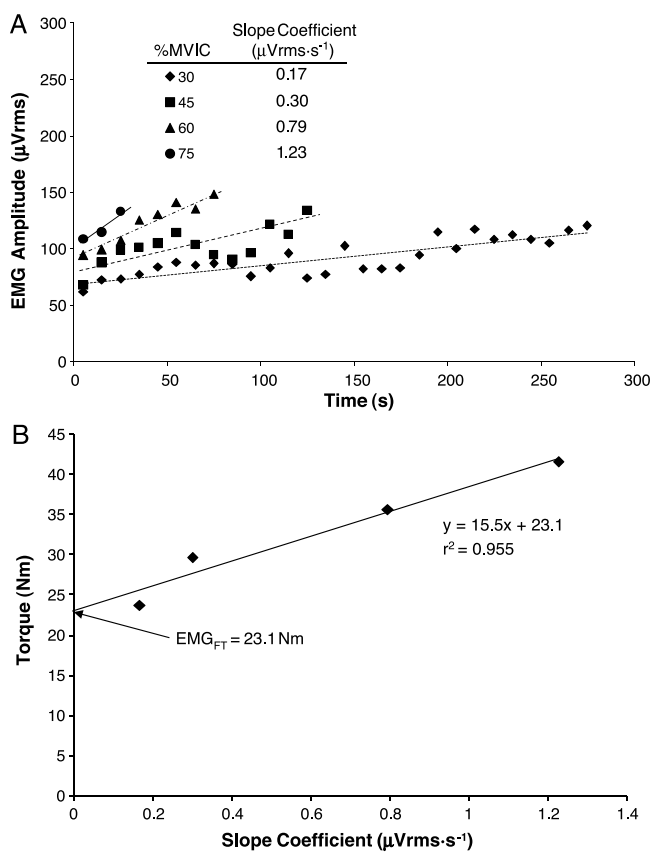

FIGURE 4—Method for estimating the EMG_{FT} from data for subject 2 for the VL. A. Each subject's EMG data (5-s epochs) were plotted as a function of time (s) for three or four percentages of MVIC work bouts. B. The slope coefficients from panel A were plotted versus torque for each of the four percentages of MVIC, and the EMG_{FT} was estimated as the *y*-intercept (slope coefficient of zero) value (23.1 N·m).

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TABLE 1. Data for each subject for the maximum voluntary isometric contraction (MVIC) and critical force (CF) test.

Subject	Gender	MVIC (N·m)	Torque (N·m)	T_{lim} (s)	r^2 (W_{lim} vs T_{lim})	CF (N·m)	CF (%MVIC)
1	F	83.0	29.0	374.3	0.994	22.1	26.6
			37.3	158.6			
			45.6	122.7			
			53.9	77.1			
2	F	59.4	23.8	301.2	0.981	17.3	29.1
			29.7	153.2			
			35.6	91.1			
			41.6	92.1			
3	M	114.7	45.9	153.7	0.913	26.7	23.3
			57.4	77.1			
			68.8	73.5			
			80.3	58.0			
4	F	113.9	34.2	216.0	0.991	14.7	12.9
			51.2	118.1			
			68.3	80.5			
			79.7	63.4			
5	F	61.8	24.7	225.9	0.863	12.1	19.5
			30.9	139.7			
			37.1	102.1			
6	M	235.9	82.6	281.1	0.953	37.1	15.7
			106.2	185.8			
			141.5	113.2			
			176.9	96.6			
7	M	235.9	70.8	257.1	0.954	29.9	12.7
			106.2	144.0			
			141.5	88.8			
8	M	276.6	83.0	217.7	0.706	47.5	17.2
			124.5	83.1			
			193.6	50.7			
Mean		147.6				25.9	17.6
SD		87.6				12.1	6.2

W_{lim} = isometric torque × time to exhaustion (T_{lim}). Due to technical problems with the EMG equipment, subjects 5, 7, and 8 completed only three continuous isometric muscle actions.

r = 0.763–0.968. When expressed as a %MVIC, the only significant correlation (r = 0.791) among the fatigue thresholds was for the EMG_{FT} for the VL and the EMG_{FT} for the VM (Table 3).

DISCUSSION

Linear relationships between W_{lim} and T_{lim} have been reported during continuous isometric muscle actions for individual muscles and synergic muscle groups (27). Monod and Scherrer (27) used these linear relationships as the basis for the CF test. The results of the present study of isometric muscle actions of the leg extensors supported those of Monod and Scherrer (27) and resulted in r^2 values for the W_{lim} versus T_{lim} relationship that ranged from 0.706 to 0.990 (Table 1). These r^2 values were similar to those previously reported for whole body analogues of the CF test, including the critical power (CP) tests for cycle ergometry (8,29), swimming (39), and kayaking (19) (r^2 = 0.982–1.000) as well as the critical velocity test for treadmill running (33) (r^2 = 0.987–0.999).

Theoretically, an isometric torque level at the CF can be maintained for an extended period without fatigue and is dependent on "circulatory conditions in the muscle" (27, p. 332), including oxygen availability for energy metabolism and blood flow (5,27,29). The mean ± SD CF for isometric leg extension in the present study was 17.6% ± 6.2% of MVIC (Table 1). Previous studies (5,27) have reported CF values during isometric muscle actions of the forearm flexors and extensors, middle finger flexors, and leg extensors that ranged from 15% to 22% of MVIC. In addition, Edwards et al. (15,16) found that for isometric muscle actions of the leg extensors, the onset of a reduction in muscle blood flow due to an increase in intramuscular pressure and occlusion of the vascular beds occurred at forces greater than 20% of MVIC. In the present study, five of the eight subjects had CF values that were less than 20% of MVIC (Table 1). Thus, the results of the present study, in conjunction with those of Monod and Scherrer (27) and Edwards et al. (15,16), suggest that, for most subjects, the T_{lim} during a continuous isometric leg extension muscle action at the CF would not be limited by a force-related restriction of muscle blood flow.

Previous studies have reported linear relationships between EMG amplitude and time during fatiguing, isometric muscle actions (3,12,13,23,28). Furthermore, the slope coefficients of EMG amplitude versus time relationships were directly proportional to the percentage of MVIC but inversely related to T_{lim} (13). The results of the present study supported these previous findings (3,12,13,23,28) and resulted in r^2 values for the EMG amplitude versus time and torque versus EMG slope coefficient relationships for the VL, the VM, and the RF muscles that ranged from 0.010 to 0.999 and from 0.637 to 0.999, respectively (Table 2). These ranges of r^2 values were also consistent with those previously reported for the EMG_{FT} test during cycle ergometry (r^2 = 0.010–0.999 and 0.030–0.999, respectively) (13,21,31). Thus, the results of the present study

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TABLE 2. Data for the vastus lateralis (VL), vastus medialis (VM), and rectus femoris (RF) muscles for each subject from the electromyographic fatigue threshold (EMG_{FT}) tests.

			VL					VM					RF				
Subject	Gender	Torque (N·m)	EMG slope (μVrms·s^{-1})	r^2 (EMG vs Time)	r^2 (Torque vs EMG Slope)	EMG_{FT} (N·m)	EMG_{FT} (%MVC)	EMG slope (μVrms·s^{-1})	r^2 (EMG vs Time)	r^2 (Torque vs EMG Slope)	EMG_{FT} (N·m)	EMG_{FT} (%MVC)	EMG slope (μVrms·s^{-1})	r^2 (EMG vs Time)	r^2 (Torque vs EMG Slope)	EMG_{FT} (N·m)	EMG_{FT} (%MVC)
1	F	29.0	0.404	0.419	0.974	22.4	27.0	0.462	0.497	0.979	15.3	18.4	—	—	0.953	16.8	20.3
		37.3	0.979	0.921				—	—				0.184	0.351			
		45.6	1.236	0.887				0.886		0.897			0.285	0.270			
		53.9	1.921	0.809				1.247	0.782				0.333	0.595			
2	F	23.8	0.166	0.710	0.955	23.1	38.8	0.134	0.811	0.819	24.6	41.5	0.018	0.016	0.764	24.3	40.9
		29.7	0.300	0.505				0.214	0.678				—	—			
		35.6	0.794	0.937				0.993	0.975				0.718	0.736			
		41.6	1.226	0.928				—	—				0.875	0.805			
3	M	45.9	0.530	0.753	0.852	47.2	41.2	2.029	0.919	0.637	34.4	30	0.287	0.470	0.705	36.4	31.7
		57.4	0.986	0.543				2.915	0.815				1.120	0.318			
		68.8	1.600	0.973				—	—				—	—			
		80.3	3.961	0.993				3.078	0.611				1.268	0.647			
4	F	34.2	0.062	0.082	0.959	34.1	29.9	0.015	0.015	0.958	37	32.4	0.053	0.032	0.922	38.7	33.9
		51.2	0.226	0.308				0.101	0.129				0.177	0.037			
		68.3	0.689	0.546				—	—				1.025	0.455			
		79.7	0.805	0.758				0.480	0.473				1.329	0.459			
5	F	24.7	0.169	0.452	0.999	17.4	28.2	0.313	0.528	0.999	11.9	19.2	0.019	0.010	0.823	26.8	43.4
		30.9	0.322	0.921				0.614	0.646				0.073	0.072			
		37.1	0.459	0.670				0.92	0.944				0.561	0.549			
6	M	82.6	0.476	0.769	0.943	36.3	20.9	0.698	0.755	0.981	26.7	20.8	0.382	0.351	0.726	55.5	43.2
		106.2	—	—				0.843	0.741				1.002	0.501			
		141.5	1.258	0.726				1.389	0.607				1.069	0.460			
		176.9	1.397	0.639				1.717	0.678				—	—			
7	M	70.8	0.809	0.760	0.990	37.8	16.0	2.389	0.815	—	38.3[a]	16.2	0.922	0.741	0.934	49.3	38.4
		106.2	1.676	0.804				2.425	0.604				1.496	0.331			
		141.5	2.168	0.894				1.709	0.889				3.038	0.886			
8	M	83.0	0.359	0.645	0.880	88.1	31.8	0.528	0.593	0.701	83.2	30.1	0.197	0.207	—	80.5[a]	29.1
		124.5	0.433	0.240				0.960	0.675				−2.192	0.186			
		193.6	2.742	0.999				2.370	0.276				0.230	0.205			
Mean						38.3	25.9				33.8	22.9				41.1	27.8
SD						22.4	8.4				22.1	8.8				20.7	8.0

[a] The EMG_{FT} for the VM for subject 7 and the RF for subject 8 were estimated using the second approximation procedure of Yates (40).

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indicated that the EMG_{FT} test, which was originally developed for cycle ergometry (13), is applicable to isometric muscle actions of the leg extensors.

The EMG_{FT}, as defined in the present study, represents the highest level of isometric torque that can be sustained without neuromuscular evidence of fatigue (26,30). The results of this study indicated, however, that the mean ± SD EMG_{FT} values for isometric leg extension were 25.9% ± 8.4%, 22.9% ± 8.8%, and 27.8% ± 8.0% of MVIC for the VL, the VM, and the RF muscles, respectively. Furthermore, 21 of the 24 individual EMG_{FT} values (Table 2) were greater than the 20% of MVIC that Edwards et al. (15,16) found coincided with the onset of a reduction in muscle blood flow (Table 2). Thus, the current findings suggested that, unlike CF, a restriction of muscle blood flow may contribute to fatigue during a continuous, isometric muscle action of the leg extensors at the EMG_{FT} for the VL, the VM, and the RF muscles.

Theoretically, the CF represents the maximal isometric torque that can be maintained for an extended period without fatigue and corresponds to the asymptote of the torque versus T_{lim} relationship (Fig. 1B). In addition, the EMG_{FT} represents the maximal torque level that can be maintained without neuromuscular evidence of fatigue. Therefore, conceptually, the CF and the EMG_{FT} should both estimate the maximal isometric torque that can be maintained continuously for an extended period without fatigue. In the present study, however, the mean EMG_{FT} value for RF was significantly greater than that of the CF (41.1 ± 20.7 and 25.9 ± 12.1 N·m, respectively). Furthermore, EMG_{FT} was greater than CF for seven of eight subjects for VL and RF as well as five of eight subjects for VM (Tables 1 and 2). These findings were consistent with those of deVries et al. (13), who reported that the EMG_{FT} was significantly greater than CP (analogous to CF in the present study) during cycle ergometry. These results were also consistent with a previous study by Pavlat et al. (32), who found that the EMG_{FT} from the VL during cycle ergometry overestimated the power outputs that could be maintained for 30 and 60 min by 42% and 52%, respectively. Thus, the results of the present study, in conjunction with those of deVries et al. (13) and Pavlat et al. (32), suggested that during isometric muscle actions of the leg extensors as well as cycle ergometry, CF (and CP) can be maintained for a longer period than the EMG_{FT}. In addition, the results of the present study indicated that there were no mean differences in EMG_{FT} values for the three superficial muscles of the quadriceps femoris (Table 2). These findings were not consistent with those of Housh et al. (21), who reported that during cycle ergometry, the mean EMG_{FT} for the RF (220 ± 30 W) was 11.3% less ($P < 0.05$) than that of the VL (248 ± 31 W). It is possible that the difference between the findings of Housh et al. (21) for cycle ergometry and those of the current study of continuous, isometric muscle actions were due to mode-specific differences in muscle blood flow. Future studies should compare the effects of continuous isometric, intermittent isometric, and dynamic muscle actions on differences in the EMG_{FT} of the VL, the VM, and the RF muscles.

The mean difference between the CF and the EMG_{FT} value for RF muscle in the present study (Tables 1 and 2) may have been due to anatomical, physiological, and/or nonphysiological factors that affected fatigue responses as well as the slope coefficients of the individual EMG fatigue curves during the isometric muscle actions (17). For example, surface EMG cannot be used to examine the fatigue responses of the vastus intermedius, and therefore, EMG_{FT} values were determined for only the VL, the VM, and the RF muscles. It is possible, however, that the vastus intermedius contributes uniquely to the fatigue responses of the quadriceps femoris muscles during continuous isometric muscle actions. Thus, determining EMG_{FT} values from only the VL, the VM, and the RF muscles may not have accurately identified the torque output associated with the threshold of fatigue for the leg extension muscle actions. Furthermore, increases in muscle temperature have been shown to attenuate EMG amplitude (34–36). Although the specific effects of changes in muscle temperature on EMG amplitude and the estimations of EMG_{FT} in the present study are unknown, Edwards et al. (15) reported a linear relationship between the force of an isometric contraction and the rate of increase in muscle temperature. Thus, it is possible that the durations of the muscle actions at the different percentages of MVIC may have affected the magnitude of the increase in muscle temperature and, therefore, the slope coefficients for the EMG fatigue curves. This, in turn, may have affected the estimations of the EMG_{FT} values in the present study. In addition, motor unit synchronization can also affect the amplitude of the EMG signal (17). Motor unit synchronization describes the tendency for active motor units to fire either simultaneously

TABLE 3. Correlations among the torque levels (N·m) and percent maximum voluntary isometric contraction (%MVIC) values for critical force (CF) and EMG fatigue thresholds (EMG_{FT}) of the vastus lateralis (VL), the vastus medialis (VM), and the rectus femoris (RF) muscles.

		CF	EMG_{FT} VL	EMG_{FT} VM	EMG_{FT} RF
Torque (N·m)	CF	1.000			
	EMG_{FT} VL	0.842*	1.000		
	EMG_{FT} VM	0.763*	0.968*	1.000	
	EMG_{FT} RF	0.872*	0.886*	0.880*	1.000
%MVIC	CF	1.000			
	EMG_{FT} VL	0.642	1.000		
	EMG_{FT} VM	0.381	0.791*	1.000	
	EMG_{FT} RF	−0.270	−0.189	0.052	1.000

* Significant at $P < 0.05$.

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or nearly simultaneously (2,11). Bigland-Ritchie et al. (4) suggested that motor unit synchronization may contribute to the fatigue-induced increase in EMG amplitude during a sustained isometric muscle action. Thus, it is possible that motor unit synchronization affected the slope coefficients for the EMG amplitude versus time relationships and, thus, the estimations of EMG_{FT} for VL, VM, and RF muscles.

Crosstalk, amplitude cancellation, and use of a bipolar electrode configuration are nonphysiological factors that may have affected the estimations of EMG_{FT} in the present study (2,17,22). Crosstalk describes a signal recorded over one muscle that is actually produced by a nearby muscle and conducted through the intervening volume to the recording electrodes (17). It is possible that crosstalk between the superficial muscles of the quadriceps femoris affected the slope coefficients for the EMG fatigue curves as well as the estimations of EMG_{FT} in the present study. Amplitude cancellation describes the attenuation of EMG amplitude due to the superposition of positive and negative phases of the action potentials of activated motor units (2,17,22). It is possible that differential effects of amplitude cancellation at the various percentages of MVIC affected the estimations of EMG_{FT}. In addition, according to Lynn et al. (24), active fibers within about 0.4 length units (1 length unit equals the interelectrode distance) tend to dominate the EMG signal. Thus, the interelectrode distance of 30 mm used in the present study suggests that the EMG_{FT} values were determined based on muscle fiber action potential responses within approximately 12 mm of the surface EMG electrodes. This small sampling area may not have represented the neuromuscular fatigue characteristics of the entire VL, VM, or RF muscles. Future studies should examine the effects of various anatomical, physiological, and nonphysiological factors on the slope coefficients for EMG fatigue curves at different levels of isometric forces as well as the estimation of EMG_{FT}.

The results of the present study indicated that there were differences in the isometric torque levels associated with fatigue thresholds estimated from the neuromuscular responses (the EMG_{FT} value from the RF muscle) and the force versus T_{lim} relationship (CF). In addition, the mean CF occurred at a torque level (17.6% MVIC) that is typically not affected by circulatory occlusion. Continuous isometric muscle actions at the EMG_{FT} (VL = 25.9%, VM = 22.9%, and RF = 27.8% MVIC), however, may be limited by restricted blood flow to the working muscles. There are several potential anatomical, physiological, and nonphysiological factors such as the inability to determine the EMG_{FT} from the vastus intermedius muscle using surface EMG, increases in muscle temperature, motor unit synchronization, crosstalk, amplitude cancellation, and bipolar electrode configuration, which may have contributed to the mean differences between the EMG_{FT} values and the CF in the present study.

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